

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology	
Subject: DSIP	Aim: Simulate cross correlation and autocorrelation on discrete time signals.	
Experiment:-3	Date:01-08-2026	Enrollment No:92400133075

Github: <https://github.com/Chiragsinh933/DSIP.git>

Objective:-

To understand the mathematical definitions of autocorrelation and cross-correlation.

To simulate these operations on deterministic and random discrete signals using Python.

To interpret the results for signal similarity, periodicity, and time-delay estimation.

Theory:-

What is Correlation?

Correlation is a mathematical way to measure how much two signals look alike as you slide one past the other to check for a match. It is widely used to find hidden patterns and calculate time delays in technologies like radar, sonar, and wireless communications.

What is Autocorrelation?

The autocorrelation function of a discrete-time signal $x[n]$ measures the similarity between the signal and a time-shifted version of itself. For a finite-length signal of length N , it is defined as:

$$r_{xx}[l] = \sum_{n=-\infty}^{\infty} x[n] \cdot x[n-l]$$

where l is the lag (integer shift).

Interpretation:

At $l = 0$, $r_{xx}[0] = \sum x[n]^2$, which is the total energy of the signal.

Autocorrelation is even symmetric: $r_{xx}[l] = r_{xx}[-l]$.

The maximum always occurs at zero lag: $|r_{xx}[l]| \leq r_{xx}[0]$.

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Cross-Correlation:-

The cross-correlation between two different signals $x[n]$ and $y[n]$ measures how well $y[n]$ matches $x[n]$ when shifted:

$$r_{xy}[l] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n-l]$$

Interpretation:-

If $y[n] = x[n-d]$ (i.e., a delayed version of x), then $r_{xy}[l]$ will have its peak at $l = d$.

Cross-correlation is not generally symmetric.

Normalized Correlation (Correlation Coefficient):-

Often we normalize to get values between -1 and 1 for easier interpretation:

$$\rho_{xy}[l] = \frac{r_{xy}[l]}{\sqrt{r_{xx}[0] \cdot r_{yy}[0]}}$$

where $\rho = 1$ means identical signals (for that lag), $\rho = -1$ means opposite polarity, and $\rho = 0$ means completely unrelated.

Applications in DSP:-

Time-delay estimation: Finding the lag where cross-correlation peaks gives the time shift between two received signals (e.g., in target ranging).

Periodicity detection: Autocorrelation of a periodic signal has peaks at multiples of the period.

Random signal analysis: White noise has an autocorrelation that is zero everywhere except at lag 0 (ideal for decorrelation).

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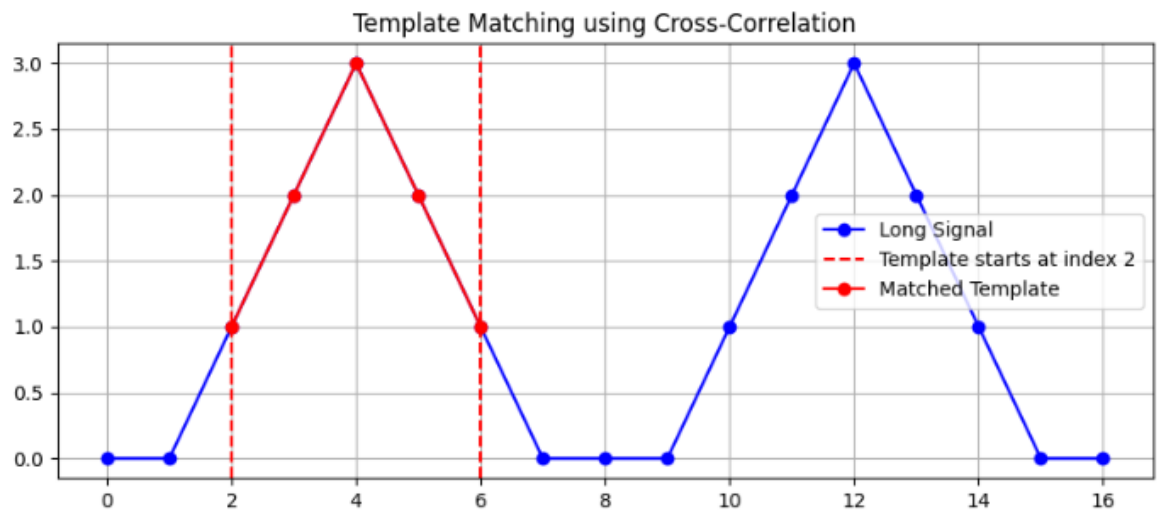
Normalized cross-correlation:-

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
# Long signal
long_signal = np.array([0, 0, 1, 2, 3, 2, 1, 0, 0, 0, 1, 2, 3, 2, 1, 0, 0])
# Template to find
template = np.array([1, 2, 3, 2, 1])
# Compute cross-correlation
corr = np.correlate(long_signal, template, mode='valid')
peak_idx = np.argmax(corr)
plt.figure(figsize=(10, 4))
plt.plot(long_signal, 'bo-', label='Long Signal')
plt.axvline(peak_idx, color='red', linestyle='--', label=f'Template starts at index {peak_idx}')
plt.axvline(peak_idx + len(template) - 1, color='red', linestyle='--')
plt.plot(np.arange(peak_idx, peak_idx + len(template)), template, 'ro-',
label='Matched Template')
plt.legend()
plt.title('Template Matching using Cross-Correlation')
plt.grid(True)
plt.show()
```

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Output:-



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Cross-Correlation to Detect Time Delay:-

Code:-

```

import numpy as np
import matplotlib.pyplot as plt
# Generate a random signal
np.random.seed(42)
x = np.random.randn(50)
# Create a delayed version (delay = 10 samples)
delay = 10
y = np.zeros_like(x)
y[delay:] = x[:-delay] # y[n] = x[n-10]
# Add a tiny bit of noise to make it realistic
y = y + 0.05 * np.random.randn(50)
# Cross-correlation
cross_corr = np.correlate(x, y, mode='full')
# Lags for cross-correlation: length = len(x) + len(y) - 1 = 99
Nx = len(x)
Ny = len(y)
lags = np.arange(-Ny + 1, Nx)
# Find the peak lag
peak_idx = np.argmax(cross_corr)
estimated_delay = lags[peak_idx]
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(x, 'b-', label='Original x[n]')
plt.plot(y, 'r--', label='Delayed y[n]')
plt.legend()
plt.title('Original and Delayed Signals')
plt.grid(True)
plt.subplot(1, 2, 2)
plt.stem(lags, cross_corr, basefmt="k-")

```

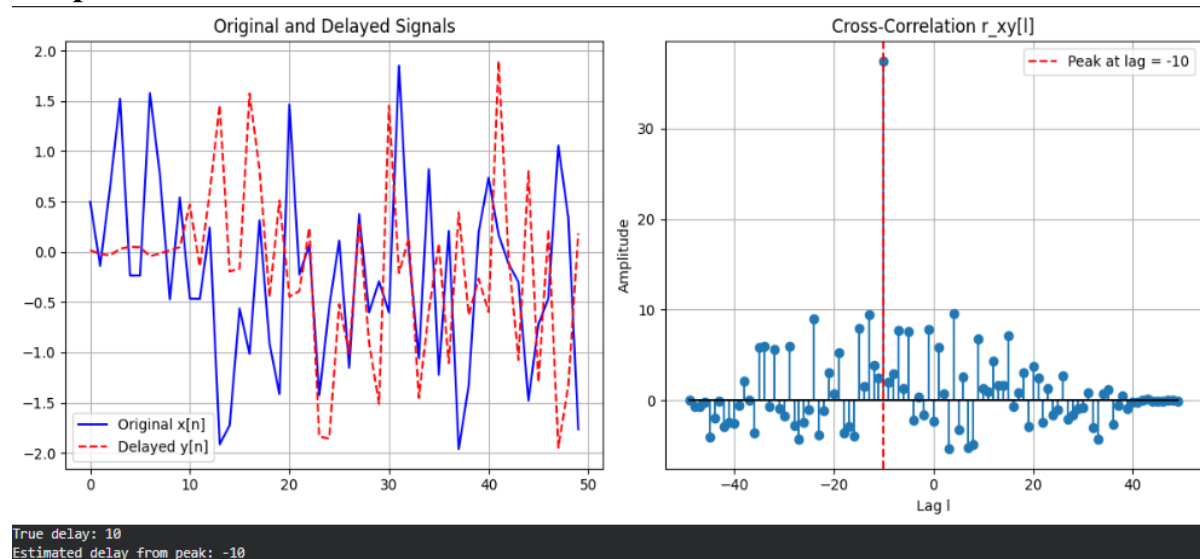
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```

plt.axvline(estimated_delay, color='red', linestyle='--', label=f'Peak at lag =
{estimated_delay}')
plt.title('Cross-Correlation r_xy[l]')
plt.xlabel('Lag l')
plt.ylabel('Amplitude')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
print(f'True delay: {delay}')
print(f'Estimated delay from peak: {estimated_delay}')

```

Output:-



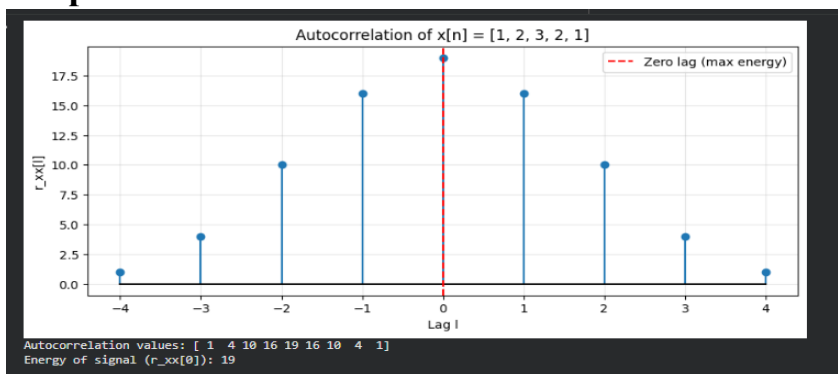
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Autocorrelation of a Deterministic Signal:-

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
# Define a simple deterministic signal
x = np.array([1, 2, 3, 2, 1])
# Compute autocorrelation using numpy.correlate (mode='full' preserves all lags)
auto_corr = np.correlate(x, x, mode='full')
# Create lag axis. For length N, lags go from -(N-1) to (N-1)
N = len(x)
lags = np.arange(-N + 1, N)
# Plot
plt.figure(figsize=(10, 4))
plt.stem(lags, auto_corr, basefmt="k-")
plt.title('Autocorrelation of x[n] = [1, 2, 3, 2, 1]')
plt.xlabel('Lag l')
plt.ylabel('r_xx[l]')
plt.grid(True, alpha=0.3)
plt.axvline(0, color='red', linestyle='--', label='Zero lag (max energy)')
plt.legend()
plt.show()
print("Autocorrelation values:", auto_corr)
print("Energy of signal (r_xx[0]):", auto_corr[N-1])
```

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Autocorrelation – Periodic vs. Random Signals:-

Code:-

```

import numpy as np
import matplotlib.pyplot as plt
n = np.arange(0, 200)
# Periodic signal: Sine wave
fs = 1
f = 0.05 # 5 cycles in 100 samples
periodic_signal = np.sin(2 * np.pi * f * n)
# Random signal (White noise)
np.random.seed(0)
noise_signal = np.random.randn(200)
# Compute autocorrelations
auto_periodic = np.correlate(periodic_signal, periodic_signal, mode='full')
auto_noise = np.correlate(noise_signal, noise_signal, mode='full')
N = len(periodic_signal)
lags = np.arange(-N + 1, N)
plt.figure(figsize=(12, 6))
plt.subplot(2, 2, 1)
plt.plot(n, periodic_signal)
plt.title('Periodic Signal (Sine)')
plt.grid(True)
plt.subplot(2, 2, 2)
plt.stem(lags, auto_periodic, basefmt="k-", markerfmt='b.')
plt.title('Autocorrelation of Periodic Signal')
plt.xlabel('Lag')
plt.xlim(-50, 50)
plt.grid(True)
plt.subplot(2, 2, 3)
plt.plot(n, noise_signal)
plt.title('White Noise')
plt.grid(True)

```

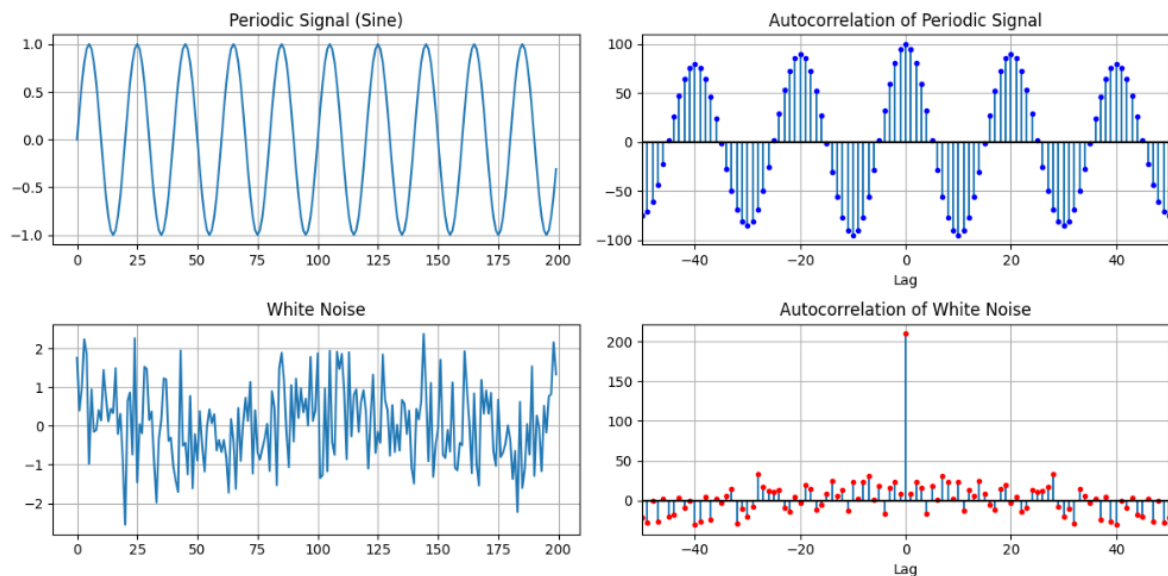

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```

plt.subplot(2, 2, 4)
plt.stem(lags, auto_noise, basefmt="k-", markerfmt='r.')
plt.title('Autocorrelation of White Noise')
plt.xlabel('Lag')
plt.xlim(-50, 50)
plt.grid(True)
plt.tight_layout()
plt.show()

```

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Conclusion:-

- **Autocorrelation:** When a signal is compared with itself, it gives a symmetrical graph with the highest peak at zero. This shows the maximum energy of the signal.
- **Finding Time Delay:** Cross-correlation helps to find the time delay between two signals by showing the highest peak at the correct position, even if there is a little noise.
- **Identifying Signals:** Periodic signals give repeated peaks, while random noise gives only one high peak at the center and small values everywhere else.
- **Pattern Matching:** Cross-correlation can easily find a small known pattern inside a longer signal, making it useful for pattern detection.